

Andrew Nguyen

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WORK EXPERIENCE

AVID

Aug. 2022 – May 2023

Tutor

Plano, Texas

- AVID is an educational organization focused on preparing students for college readiness and success.
- Led multiple students simultaneously to improve academic performance by implementing tailored learning strategies.

EDUCATION

University of Texas at Dallas

Aug. 2022 - Dec. 2025

Bachelor of Science in Computer Science

Richardson, Texas

FEATURED PROJECTS

Portfolio Website | HTML, CSS, JavaScript

July 2024 (Updated Jan 2026)

- General static web page using HTML, CSS, and JavaScript to showcase projects and skills
 - Built with semantic HTML5, custom CSS for responsive layouts, and JavaScript for interactivity.
 - Deployed on GitHub Pages, enabling a fast, reliable, and cost-free public portfolio.

URL Shortener (go.a04.dev) | NextJS, NeonDB, TailwindCSS, TypeScript

Dec. 2025 - Jan. 2026

- Self-hosted URL shortener built with Next.js and Postgres, featuring expiring links, browser extensions, and a secure, authenticated API. Currently only published to Chrome Web Store @[URL Shortener - go.a04.dev](https://chrome.google.com/webstore/detail/url-shortener-go-a04-dev)
 - Implemented bearer-token auth, per-IP/per-token rate limiting, and SSRF protection to harden the redirect/shorten pipeline.
 - Delivered Chrome/Edge + Firefox extensions with QR code generation and an admin CLI for API key and link lifecycle management (create/list/revoke).

Homelab Media/Game Server | Proxmox, OMV, Docker Compose, Wireguard, NPM

Dec. 2025 - Developing

- Building a homelab on Proxmox with VM isolation, shared storage, and minimal WAN exposure, currently on VM and will migrate to baremetal. Will be used for a testing environment in the future.
 - Set up a 3-VM layout (OMV/media, infra, games) with raw HDD passthrough to OMV and a TRaSH-style /data structure for hardlinks.
 - Routed qBittorrent through Gluetun (Proton WireGuard) with qSticky port-sync, and enabled remote access via WireGuard.
 - Deployed services with Docker Compose and added backups/restore tests plus secure secrets and firewall defaults.

TMDB Movie Database Web App | Next JS, PostGRES, Prisma, TypeScript

Oct. 2025 – Developing

- Building a movie watchlist web app with Next.js 15, TypeScript, and Tailwind CSS, using TMDB API for movie data and Prisma + PostgreSQL for secure, persistent watchlists with Supabase Auth.
 - TMDB API integration for server-side movie search, details, and watchlist management.
 - User authentication and watchlist storage via Supabase Auth and Prisma ORM.

Human Pose Detection | Python, OpenCV, MediaPipe

Nov. 2025

- Developed a real-time human pose and gesture recognition system using Python, OpenCV, MediaPipe, and DeepFace, capable of detecting skeletal landmarks, counting fingers, and identifying facial emotions.
 - Real-time pose tracking with MediaPipe Pose and skeletal visualization.
 - Emotion detection via DeepFace with GPU acceleration for fast, real-time analysis.

CERTIFICATIONS, SKILLS & TECHNOLOGIES

- **Skills:** Communication Skills; Project Management; Software Development; Agile Methodologies
- **Technologies:** HTML; CSS; JavaScript; C++; Python; Node.js; React; NextJS; Linux; GitHub; Git; Docker; PostgreSQL, MongoDB, Supabase, Convex, Firebase